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ORIGINAL ARTICLE

Genome-wide identification, characterization analysis and expression profiling of auxin-responsive GH3 family genes in wheat (*Triticum aestivum* L.)

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Abstract

Auxin affects many aspects of plant growth and development by regulating the expression of auxin-responsive genes. As one of the three major auxin-responsive families the Gretchen Hagen3 (GH3) gene family maintains hormonal homeostasis by conjugating excess indole-3-acetic acid (IAA), salicylic acid (SA), and jasmonic acid (JA) to amino acids during hormone and stress-related signaling. Although some work has been carried out the functions of wheat GH3 (TaGH3) family genes in response to abiotic stresses (including salt stress and osmotic stress) are largely unknown. Access to the complete wheat genome sequence permits genome-wide studies on TaGH3s. We performed a systematic identification of the TaGH3 gene family at the genome level and detected 36 members on 14 wheat chromosomes. Many of the genes were segmentally duplicated and Ka/Ks and inter-species synthetic analyses indicated that polyploidization was the contributor to the increased number of TaGH3 members. Phylogenetic analyses revealed that TaGH3 proteins could divided into three major categories (TaGH3-I, TaGH3-II, and TaGH3-III). Diversified *cis*-elements in the promoters of *TaGH3* genes were predicted as essential players in regulating TaGH3 expression patterns. Gene structure and motif analyses indicated that most *TaGH3* genes have relatively conserved exon/intron arrangements and motif compositions. Analysis of multiple transcriptome data sets indicated that many *TaGH3* genes are responsive to biological and abiotic stresses and possibly have important functions in stress response. qRT-PCR analysis revealed that *TaGH3s* were induced by salt and osmotic stresses. Customized annotation results revealed that TaGH3s were widely involved in phytohormone response, defense, growth and development, and metabolism. Overall, our work provides a comprehensive insight into the TaGH3 family members, and a basis for the further study of their biological functions in wheat.

Keywords Expression profiles · Genome-wide analysis · GH3 · Phylogenetic analysis · qRT-PCR · Synteny analysis

Introduction

As the first plant hormone discovered, auxin plays a key role in the entire growth and development of plants. Auxin can induce auxin primary response genes and rapid high expression.[1]. It participates in physiological processes including plant apical dominance, root formation, reproduction, and response to biotic and abiotic stimuli [2, 3]. Phytohormones are involved in many distinct and/or overlapping processes throughout the life cycle of plants [4]. The auxin early-response genes could be roughly divided into three categories: Aux/IAAs (Auxin/indoleacetic acid), GH3s (Gretchen Hagen 3), and SAURs (Small auxin up RNAs). Auxin can induce rapid and transient expression of most genes in these three gene families, thereby promoting plant resistance to

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environmental stress.[5, 6]. Since GH3 proteins are involved in maintenance of auxin homeostasis, it is imperative to have a better understanding of their function during growth and development, and the mechanism of auxin action in these processes [7].

The GH3 gene family is a major class of primary auxin response genes. The first GH3 gene described in 1987 was isolated from soybean (*Glycine max*) as an early auxin-responsive gene [8]. GH3 gene family members were found later in plants such as *Nicotiana tabacum* [9], *Arabidopsis thaliana* [10], *Oryza sativa* [11] and others [12–14]. GH3 proteins were also identified in mosses, *Physcomitrella patens* [15], and *Marchantia polymorpha* [16] which predated land plants by nearly 400 million years. GH3 proteins have been implicated in several developmental processes in plants. Two Arabidopsis GH3 proteins namely DFL1 and YDK1 were reported to negatively regulate shoot cell elongation and lateral root formation [17, 18]. Khan and Stone (2007) showed that At GH3.9 affected the development of primary roots of plants by regulating the activity of auxin [19]. Chen et al. (2010) provided insights into reactions catalyzed by GH3 family enzymes to modulate plant hormone action [20]. They reported that rice GH3-8 protein requires either Mg^{2+} or Mn^{2+} for maximum activity.

The promoters of plant GH3 genes usually contain auxin response elements (AuxRE, TGTCTC). AuxREs motifs can vary in number and sequence (TGTCCC, TGTCAC) [21, 22]. Members of another gene family named auxin response factor (ARF) regulate expression of the auxin-responsive genes [1, 23]. GH3 family genes are divided into three subfamilies (I, II, and III) based on sequence similarities and substrate specificities of their transcripts in Arabidopsis, which has 19 GH3 members and one incomplete *GH3* gene [24, 25]. Sub-family I GH3 enzymes are jasmonic acid (JA)-amido or salicylic acid (SA)-amido synthetases [24]. JA is vital for defense. It is synthesized within minutes to induce the expression of many genes and affects reaction to herbivory and disease through the JA pathway [26]. Arabidopsis sub-family II GH3 proteins were demonstrated to be active on IAA-amido synthetases [10]; and *AtGH3.5* (*AtGH3a*) can also catalyze the adenylation of SA and the linkage with amino acids. The phytohormone IAA is responsible for growth and development in flowering plants. It influences cell division, expansion, and differentiation GH3 in subfamily III were Arabidopsis-specific enzymes [24, 27, 28]. *AtGH3.12* (*PBS3*) belongs to the GH3 III sub-family and is thought to catalyze the coupling reaction between SA and amino groups [29]. And studies have found that *Arabidopsis GH3.15* (*AtGH3.15*) preferentially uses indole-3-butyric acid (IBA) and glutamine as substrates. AtGH3.15 is unique in the GH3 protein family because it modifies the role of IBA in auxin homeostasis, and it is the first GH3 protein that primarily modifies plant growth regulators other than IAA and JA. The functions of other members have not been determined.

As an important auxin responsive gene family, *GH3* genes play roles in the interaction between auxin homeostasis and growth regulation with stress adaptation responses [30]. For example, *OsGH3.2*, a GH3 family member in rice, affects cold and drought tolerance by modulating abscisic acid content [31]. Another example is *OsGH3.1* inhibition of cell growth and cell wall loosening enhances rice resistance to fungal pathogen and overexpression of *OsGH3.1* accelerates this process by reducing auxin content [32]. Activation of *OsGH3.13* reduces the auxin content to enhance drought tolerance in rice and alters root architecture [33]. Treatment with SA stabilized Aux/IAAs inhibitory proteins and completely inhibited the expression of auxin-responsive genes, suggesting that SA inhibition of auxin signaling could be a step in SA-mediated disease resistance [34]. Over-expression of the *AtGH3.5* gene enhances plant systemic disease resistance and suggests that the *GH3* gene mediates interaction of SA with the auxin signaling pathway [35]. Overall, all of these reports suggested that GH3-mediated auxin homeostasis is an important constituent of auxin signaling in response to stress.

Recently, the GH3 gene family has been identified in different species, including moss (*Physcomitrella patens*), tomato (*Solanum lycopersicum*) and apple (*Malus domestica*) [13, 15, 36]. Taken together, the above evidence suggests that several *GH3* genes were clearly implicated in the regulation of plant stress resistance, an agronomical characteristic of great relevance. Wheat (*Triticum aestivum*) is one of the three major world crops [37]. However, there is a lot more that is yet to be discovered in the hexaploid TaGH3 family. And the comprehensive information and expression patterns of GH3 family genes responses to abiotic stresses in wheat were largely unknown. It is of great significance for scientific research. The results of this study provided the basis for further study of wheat GH3 gene and provided some candidates with potential application in molecular breeding to improve crop stress tolerance.

Materials and methods

Identification and classification the GH3 gene family members in wheat

Twenty *Arabidopsis* GH3s (AtGH3s), 12 maize GH3s (ZmGH3s) and 16 rice GH3s (OsGH3s) protein sequences were retrieved from the Arabidopsis Information Resource (TAIR10) database (https://www.arabidopsis.org/index.jsp), the Maize Genetics and Genomics Database (MaizeGDB) (https://www.maizegdb.org/), and the Rice Genome Annotation Project (RGAP) database (https://rice.plantbiolo gy.msu.edu/), respectively. The genome sequences of wheat were downloaded from The International Wheat Genome Sequencing Consortium website (IWGSC RefSeq v1.1, https://wheat-urgi.versailles.inra.fr/Seq-Repository/Assemblies) [38]. We used AtGH3, ZmGH3, OsGH3 protein sequences as query sequences in a search of wheat protein sequences using a local BLASTp search, accepting those with an E-value of $< e^{-5}$ as TaGH3 proteins. We used a profiled Hidden Markov Model (HMM) implemented with default parameters in HMMER v3.2.1 to search for TaGH3 proteins with the GH3 domain (PF03321) of zf-GH3 in the Pfam database. Finally, we determined by Pfam whether an obtained sequence contained a GH3-specific structural conserved domain and finally determined the number of TaGH3 gene family members [39].

Analysis of conserved TaGH3 motifs and gene structures

Annotation information of *TaGH3* genes was interpreted using GSDS V2.0 (https://gsds.cbi.pku.edu.cn/index.php) to produce *TaGH3* gene structure, intron/exon distribution, and intron/exon boundaries [40]. Conserved TaGH3 gene sequences were identified using a MEME suite analysis (V4.9.1) and MAST Primer Search (https://meme-suite.org/tools) software tools [41]. The parameters were established using known GH3 protein sequences, including AtGH3s, OsGH3s, and ZmGH3s, and were then applied to identify conserved TaGH3s as follows: each sequence could comprise any number of non-overlapping occurrences of each motif, the number of different motifs was 20, and motif length ranged from 6 to 50 amino acids. The functions of these predicted motifs were analyzed using InterPro (https://www.ebi.ac.uk/interpro) and SMART (https://coot.embl-heidelberg.de/SMART), and TBtools software (https://github.com/CJ-Chen/TBtools) was used for graphical visualization [42, 43].

In silico characterization of predicted TaGH3 proteins

Characterization of TaGH3s was performed using protein identification and analysis tools on the ExPASy server10 (https://prosite.expasy.org/) [44]. The characteristics of protein length, isoelectric point (pI), molecular weight (MW), instability index, atomic composition, grand average hydropathicity (GRAVY) and amino acid composition were predicted. The online tools TMHMM (https://www.cbs.dtu.dk/services/TMHMM/) and SignalP4.1 (https://www.cbs.dtu.dk/services/SignalP/) were used to predict transmembrane domains and signal peptides in TaGH3s [45]. Subcellular localization prediction of TaGH3s was performed using Plant-mPLoc (https://www.csbio.sjtu.edu.cn/cgi-bin/PlantmPLoc.cgi) [46]. TaGH3 members were modeled three-dimensionally using the Phyre^2 (https://www.sbg.bio.ic.ac.uk/phyre2/html/) server in intensive mode [47]. Structure validation was conducted using Ramachandran plots (https://www.mordred.bioc.cam.ac.uk/rapper/rampage.php/) [48].

Sequence alignment and phylogenetic analysis

GH3 protein sequences from *Arabidopsis*, maize, rice and wheat from TAIR10, MaizeGDB, RGAP, and IWGSC, respectively. ClustalW2 was used to align all of the amino acid sequences based on the amino acid sequences of 20 AtGH3s, 12 ZmGH3s, 16 OsGH3s, and 36 TaCH3s, [49]. Phylogenetic relationships were inferred using the Maximum Likelihood (ML) method based on an LG model in MEGA7.0 software [50]. A midpoint rooted base tree was produced using the Interactive Tree of Life (IToL, version3.2.317, https://itol.embl.de).

Chromosomal locations

The wheat genome GFF3 gene annotation file was obtained from the wheat database IWGSC v1.1 and gene annotation of wheat TaGH3s were extracted from the GFF3 file. The start and end locations of TaGH3s in corresponding chromosomes was used to produce a physical map using "MapInspect" software [51].

Gene duplication and Ka/Ks (nonsynonymous to synonymous mutation rates) analyses

Gene duplications were divided into tandem and segmental replications. According to definitions in related research a tandemly replicated gene was defined as 2–5 separate paralogous genes on a single chromosome, mostly in the form of a gene cluster. A segmentally duplicated gene refers to a paralogous gene located in the same chromosomal replication region [52]. Duplications were detected according to reported methods [53]. The Ks value and Ka/Ks ratio were calculated after identification of a duplicated gene, and the selection pressure and selection mode were analyzed. The CDS (coding sequences) of the duplicated gene pairs were compared using MEGA 7.0 software. After removal of gaps the alignment results were imported into DnaSP v6.12 software for Ks value and Ka/Ks ratio analyses [54]. For estimation of the timing of duplication events the formula $T = Ks/2\lambda \times 10^{-6}$ Mya was used to calculate divergencde time (T) in millions of years (Mya), where $\lambda = 6.5 \times 10^{-9}$ represented the rate of replacement of each locus per herb plant year [52, 55]. The reference genomes of *Aegilops tauschii* (DD), *T. urartu* (AA), and *T. dicoccoides* (AABB) were download from NCBI, and *GH3* gene identification followed the same procedure as for *TaGH3* genes. Duplicated gene pairs between species were identified and used to carry out an inter-species synthetic analysis using R package "circlize" [56].

Analysis of hormone and stress-related *cis*-regulatory element

To survey *cis*-regulatory elements in the promoter regions of *TaGH3* genes 1 to 1.5 kb segments upstream of the transcription start codons were scanned for auxin and stress-related *cis*-regulatory elements. The sequences data of *TaGH3* promoters were obtained from IWGSC RefSeqv1.1 searched for *cis*-regulatory elements by the CARE program (https://bioinformatics.psb.ugent.be/webtools/plantcare/html/) in the PlantCARE database [57].

GO annotation and expression pattern profiling of TaGH3s

Multiple databases, including GO (Gene Ontology), were used to perform functional annotation of *TaGH3* genes. The full-length protein sequences were uploaded into the program, and the NCBI database was chosen as the reference to analyze molecular function, cellular components, and biological processes [58].

Original RNA-seq data from multiple transcriptome data sets, including numerous development stages and tissues from variable treatments, were download from the NCBI data base and mapped to the wheat reference genome using hisat2. Genes were then assembled using Cufflinks to assess expression levels of *TaGH3s* (normalized fragments per kilobase of the exon model per million mapped reads). The R package "pheatmap" was used to produce a heatmap of *TaGH3s*.

Growth and stress treatment of wheat seedlings

Seeds of common wheat cultivar (cv) Emai 170 were surface sterilized with 1% hydrogen peroxide, thoroughly rinsed with distilled water, and germinated for 2 days in a 20 °C incubator [59]. Seedlings were transferred and cultured in a continuously ventilated 1/4 strength Hoagland's nutrient solution [60], which was increased to 1/2 strength after 3 days. Three days later the plants were treated with 150 mM sodium chloride (NaCl), 20% polyethylene glycol (PEG) and 180 mM mannitol. The pH of the nutrient solution was adjusted to 6.0 every 2 days using 1 M KOH or 0.2 M H_2SO_4. Conditions during the treatment were 26 °C and 16 h/8 h (day/night) photoperiod. Three biological replicates were set per treatment. Leaves and roots were collected at 2, 4, 8, 12, 24, 72, 120 h post treatment using three pooled seedlings. Each treatment consisted of three replicates, each of which included three technical replicates. Samples were immediately frozen with liquid nitrogen and stored at − 80 °C.

Real-time quantitative PCR and data analysis

To elucidate developmental and tissue-specific expression profiles of *GH3* genes in wheat, quantitative real-time PCR (qRT-PCR) was performed to detect the expression level of the *GH3* gene. Total RNAs from different tissues (leaves and roots) were reverse transcribed with a 5X All-In-One RT Master Mix (Perfect Real Time) kit (ABM, Canada) into cDNAs for qPCR analysis. Gene-specific primers were designed using Primer Premier 5.0. For qRT-PCR assays, the cDNA was diluted to 400 ng/μL with ddH_2O. The reaction system contained 5 μL of 2 × SYBR Green Mix, 0.5 μL of each primer (10 μM), 0.5 μL of template (about 400 ng/μL), and 3.5 μL of ddH_2O to make a total volume of 10 μL. The protocol was carried out as follows: pre-denaturation at 94 °C for 3 min (step 1), denaturation at 94 °C for 10 s (step 2), primer annealing/extension and collection of fluorescence signal at 60 °C for 30 s (step 3). The next 40 cycles started at step 2. Three biological replicates were performed for each sample, with three technical replicates repeated each. Relative expression levels were determined using the $2^{-\Delta\Delta Ct}$ method [61]. According to research by Paolacci et al. [62] stable ADP-ribosylation factor Ta2291 (forward: GCT CTCCAACAACATTGCCAAC, reverse: GCTTCTGCC TGTCACATACGC) expressed under various conditions was used as an internal reference gene for qRT-PCR analysis. The expression level of *TaGH3* genes were plotted using Origin software [62].

Results

Identification of the GH3 gene family in common wheat

Genome-wide searches were carried out using both the Hidden Markov Model (HMM) and BLASTp with 20 *GH3* genes from *Arabidopsis*, 12 *GH3* genes from maize, and 16 *GH3* genes from rice as queries to identify *GH3* family members in wheat. Thirty six non-redundant *GH3* genes were identified (Table 1). All amino acid and full-length cDNAs *TaGH3*s are listed in Supplemented Table Sl. The sequences were named in ascending order based on phylogenetic relationship [10]. Multiple sequence alignment analyses suggested that all identified *TaGH3*s contained a typical *GH3* domain. All multi-sequence results are provided in Fig. S1. Alternative splicing isoforms were observed in *TaGH3-1.7*, *TaGH3-2.2*, *TaGH3-2.4*, and *TaGH3-2.5* (Table 1).

Table 1 Predicted sequence features of TaGH3s

Group	Designation	Gene ID	[a]Length	[b]MW	[c]pI	[d]Ins	[e]Ali	[f]GRAVY	[g]Sub
GH3-I	TaGH3-1.1	TraesCS2A02G115000.1	590	66.85	5.44	42.64	92.95	−0.153	[h]Chl
	TaGH3-1.2	TraesCS2B02G134200.1	624	70.74	5.66	44.53	95.38	−0.124	Chl
	TaGH3-1.3	TraesCS2D02G115800.1	590	66.78	5.43	42.38	94.44	−0.149	Chl
	TaGH3-1.4	TraesCS4A02G192200.1	602	66.34	5.38	44.11	92.71	−0.074	Chl
	TaGH3-1.5	TraesCS4B02G123200.1	603	66.76	5.39	47.98	92.4	−0.069	Chl
	TaGH3-1.6	TraesCS4D02G121300.1	609	67.56	5.51	46.19	92.13	−0.09	Chl
	TaGH3-1.7a	TraesCS7A02G064700.1	511	56.99	5.48	46.17	85.15	−0.082	Chl
	TaGH3-1.7b	TraesCS7A02G064700.2	622	69.44	5.5	47.55	86.74	−0.135	Chl
	TaGH3-1.7c	TraesCS7A02G064700.3	624	69.7	5.5	46.89	86.92	−0.13	Chl
	TaGH3-1.8	TraesCS7D02G059300.1	626	69.62	5.51	43.69	87.75	−0.141	Chl
GH3-II	TaGH3-2.1	TraesCS1A02G068900.1	196	21.8	9.15	62.11	84.59	−0.155	Chl. [i]Nuc
	TaGH3-2.2a	TraesCS1A02G425100.1	624	69	5.33	37.59	88.48	−0.195	Chl
	TaGH3-2.2b	TraesCS1A02G425100.2	616	68.57	5.8	43.37	89.43	−0.203	Chl
	TaGH3-2.3	TraesCS1A02G440000.1	197	22.14	9.45	69.33	86.14	−0.201	Nuc
	TaGH3-2.4a	TraesCS1B02G459500.1	621	68.6	5.44	35.98	89.69	−0.166	Chl
	TaGH3-2.4b	TraesCS1B02G459500.2	625	69.07	5.72	37.31	90.35	−0.108	Chl
	TaGH3-2.5a	TraesCS1D02G434100.1	625	68.97	5.35	36.95	89.28	−0.16	Chl
	TaGH3-2.5b	TraesCS1D02G434100.2	673	75.01	6.84	42.76	88.54	−0.169	Chl
	TaGH3-2.6	TraesCS3A02G145300.1	589	63.96	6.3	42.81	85.72	−0.036	Chl
	TaGH3-2.7	TraesCS3B02G172400.1	589	64.13	5.96	44.4	86.18	−0.033	Chl
GH3-III	TaGH3-3.1	TraesCS1A02G320200.1	617	67.63	5.92	47.59	86.5	−0.093	Chl
	TaGH3-3.2	TraesCS1B02G332400.1	620	68.04	6.12	47.83	86.24	−0.101	Chl
	TaGH3-3.3	TraesCS1D02G320000.1	621	68.08	6.07	47.52	86.1	−0.108	Chl
	TaGH3-3.4	TraesCS2A02G137800.1	392	43.1	9.31	53.78	71.3	−0.354	Chl. [j]Mit
	TaGH3-3.5	TraesCS2A02G183900.1	615	67.69	5.9	45.34	88.57	−0.087	Chl
	TaGH3-3.6	TraesCS2B02G161700.1	601	65.9	5.71	45.18	86.86	−0.147	Chl
	TaGH3-3.7	TraesCS2B02G210600.1	614	67.71	5.79	45.98	88.55	−0.1	Chl
	TaGH3-3.8	TraesCS2B02G600800.1	366	39.5	9.81	74.07	75.3	−0.241	Nuc
	TaGH3-3.9	TraesCS2D02G140800.1	283	31.14	8.85	40.99	87.56	−0.14	Chl
	TaGH3-3.10	TraesCS2D02G191800.1	615	67.71	5.79	46.52	88.57	−0.095	Chl
	TaGH3-3.11	TraesCS3A02G301200.1	611	67.69	5.49	46.91	85.63	−0.138	Chl
	TaGH3-3.12	TraesCS3A02G324100.1	613	67.64	5.69	52.85	87.05	−0.115	Chl
	TaGH3-3.13	TraesCS3B02G335300.1	612	67.82	5.42	45.89	86.11	−0.127	Chl
	TaGH3-3.14	TraesCS3B02G353200.1	613	67.7	6.05	50.82	88.01	−0.115	Chl
	TaGH3-3.15	TraesCS3D02G300600.1	611	67.68	5.56	46.12	86.12	−0.126	Chl
	TaGH3-3.16	TraesCS3D02G317600.1	614	67.87	5.85	53.04	86.74	−0.139	Chl

[a]Length (Amino acid length)
[b]MW (Molecular weight, KD)
[c]pI (Isoelectric point)
[d]Ins. (Instability index)
[e]Ali. (Aliphatic index)
[f]GRAVY (Grand average of hydropathy)
[g]Sub. (Subcellular localization)
[h]Chl. (Chloroplast)
[i]Nuc. (Nucleus)
[j]Mit. (Mitochondria)

Phylogenetic analysis divided TaGH3s into three sub-groups

A phylogenetic tree was generated using the available full-length amino acid sequences using the Maximum Likelihood (ML) method to gain better understanding of the evolutionary history and evolutionary relationships of the GH3 gene family in wheat and other species. Twenty AtGH3 proteins [10], 12 OsGH3 proteins [11], 12 ZmGH3 proteins [63], and 36 TaGH3 proteins separated into three-groups (I, II, and III) (Fig. 1) in a similar pattern to previously reported results [10, 63]. An ML phylogenetic tree was constructed to determine the phylogenetic relationship of the GH3 family within common wheat. Again the 36 TaGH3s gathered into three groups. The GH3-I group includes 10 from TaGH3s (TaGH3-1.1 to GH3-1.8), 11 from AtGH3s, 3 from OsGH3s, and 2 from ZmGH3s. Similarly, the GH3-II group includes 10 TaGH3s (TaGH3-2.1 to TaGH3-2.7), 6 OsGH3s, 5 ZmGH3s, and 3 AtGH3s. GH3-III contains 16 TaGH3s (TaGH3-3.1 to TaGH3-3.16), 7 OsGH3s, 5 ZmGH3s, and 6 AtGH3s. The names of TaGH3 genes are renamed according to their clustering groupings in the phylogenetic tree. The locus ID and corresponding gene names are listed in Table 1 and Supplemental Table S1. Predictably the relationship between wheat and rice was closer than that between wheat and Arabidopsis or maize. Members of the same family tended to have similar functions, so the grouping in the phylogenetic tree likely reflected differentiation of gene function during evolution.

Protein feature and predicted subcellular localization analysis

The amino acid sequences of the 36 TaGH3s proteins were submitted to the ExPASy Server10 (https://www.expasy.org/tools/) online analytical system for analysis of protein properties such as isoelectric point (pI), relative molecular mass (MW), and instability index (Table 1). The sizes of the deduced TaGH3 proteins varied from 196 (TaGH3-2.3) to 673 (TaGH3-2.5b) amino acids, the corresponding molecular masses varied from 21.80 to 75.01 kDa, and the predicted average pIs varied from 5.33 (TaGH3-2.2a) to 9.81 (TaGH3-3.8). Instability indices indicated that all GH3-I and GH3-III sub-family proteins were unstable. Most of

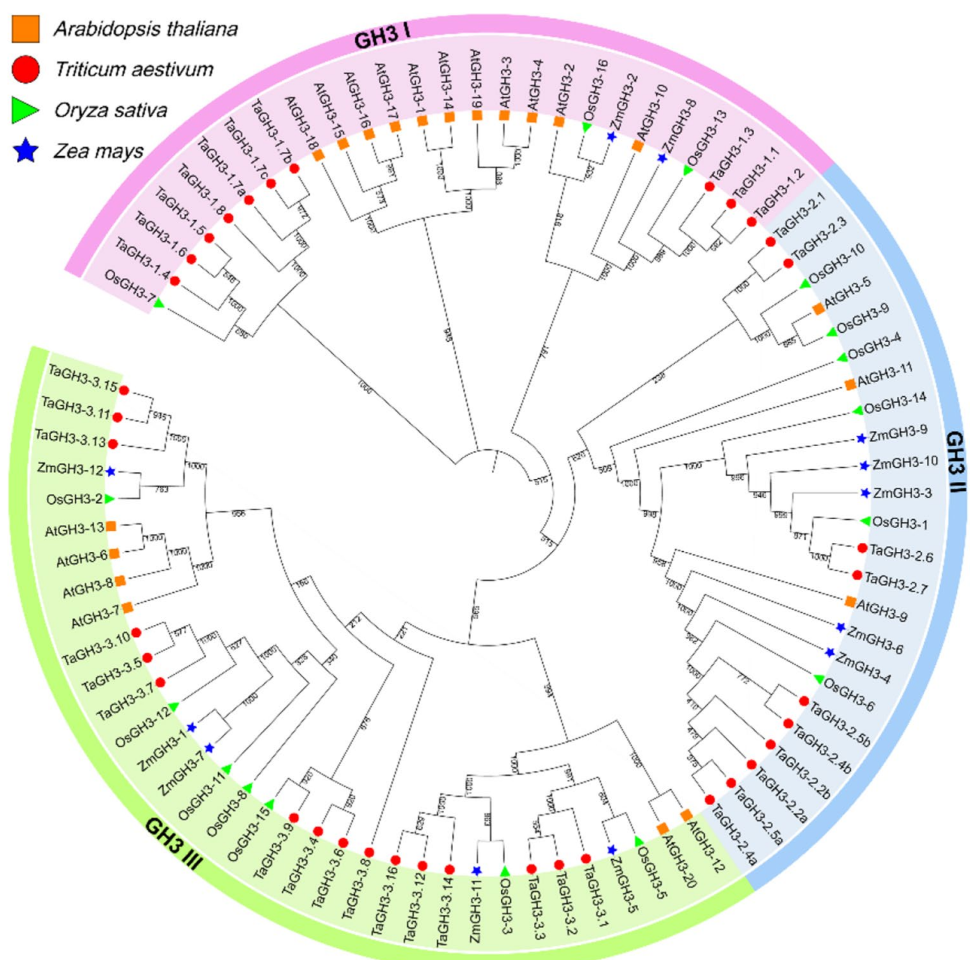


Fig. 1 Phylogenetic relationship of TaGH3s, OsGH3s, AtGH3s, and ZmGH3s. Protein sequences were aligned using ClustalW2 sequence alignment program and the phylogenetic tree was constructed by software MEGA7 used to create maximum likelihood (ML) under the LG model. The tree was constructed with 1000 bootstrap replications. Different groups were marked by different colors, and the GH3 from wheat, rice, maize and Arabidopsis were distinguished with different color and shape. (Color figure online)

the GH3-II proteins were also unstable, except for TaGH3-2.2a, TaGH3-2.4a, TaGH3-2.4b, and TaGH3-2.5a. GRAVY analysis showed that all TaGH3 proteins were hydrophilic (Table 1).

Subcellular localizations predicted using Plant-mPLoc software showed that the majority of TaGH3 proteins were in the chloroplast, and only three were localized to the nucleus (TaGH3-2.1, TaGH3-2.3, and TaGH3-3.8) and mitochondrion (TaGH3-3.4) (Table 1).

Analyses of TaGH3 gene structure and composition of conserved motifs

As indicated above TaGH3s clustered into three sub-groups highly consistent with grouping in Arabidopsis [10]. Exon–intron structural diversity often plays a key role in the evolution of gene families and can provide additional evidence to support phylogenetic grouping [64, 65]. The exon–intron structure of TaGH3 genes were detected according to their evolutionary classification. As shown in Fig. 2b, *TaGH3*s contained one to five introns, and most had a similar intron phase distribution. Seven of the 10 TaGH3-I genes had three introns; *TaGH3-1.1* and *TaGH3-1.2* contained five and TaGH3-*1.4* contained four. The number of introns in TaGH3-II genes ranged from zero to four. *TaGH3-2.2a/b*, *TaGH3-2.4a/b*, and *TaGH3-2.5a/b* had four introns; and *TaGH3-2.6* and *TaGH3-2.7* had three introns; *TaGH3-2.1* contained only one intron; *TaGH3-2.3* has no intron. Nine TaGH3-III had one intron, 6 had two, and *TaGH3-3.8* had no intron. Generally, the numbers and lengths of exons and introns were specific for each subfamily (Fig. 2b). The divergent gene structures among the different phylogenetic subgroups suggest that the genes evolved into diverse exon–intron structures to accomplish different functions.

Prediction of protein motifs is a useful method of protein analysis. To further examine structural diversity and predict the function of TaGH3s protein, 20 motifs were identified by MEME software and further annotated using InterProScan5 (Fig. 2c). Details of these 20 motifs are shown in Supplementary Table S2. The lengths of these motifs varied between 8 (motif 18) and 50 (motifs 2 and 3) amino acid residues. The number of motifs in each TaGH3 protein varied from 3 to 19; 14 motifs were functionally defined. Motifs 1 to 14 together constitute the GH3 conserved sites. The TaGH3 protein family members have highly conserved motifs. All TaGH3s in the GH3-I sub-family contain motifs 1 to 6, 8, 11, and 12. In the TaGH3-II sub-family, all members contain motifs 5 and 12. Motifs

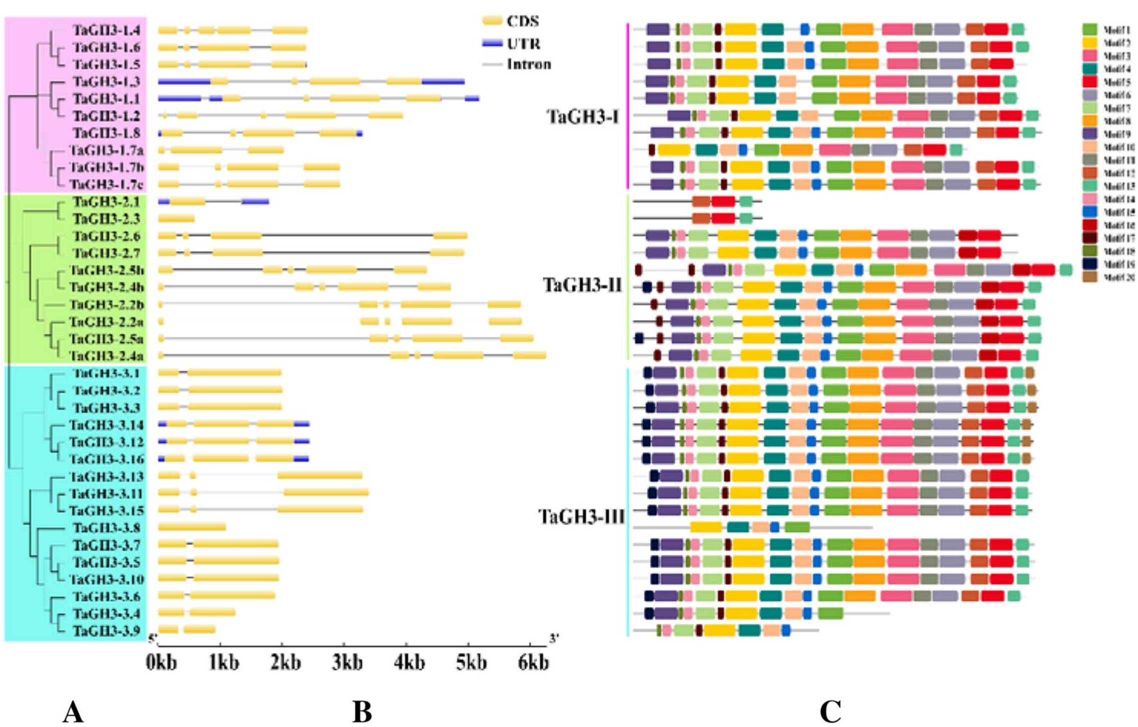


Fig. 2 Phylogenetic analysis, gene structure, and conserved motifs of TaGH3s. **a** The phylogenetic tree of all TaGH3 genes. The tree was created with a bootstrap of 1000 replications by the maximum likelihood (ML) method in MEGA7. **b** Exon–intron structure of GH3 genes in common wheat. Exon–intron structure analyses were conducted using the GSDS database. Lengths of exons and introns of each TaGH3 gene are displayed proportionally. Untranslated regions (UTRs) are indicated by blue boxes; exons are indicated by yellow boxes; introns are indicated by black lines. **c** Motif compositions were identified by MEME. Model exhibition of motif compositions in GH3 amino acid sequences were prepared using MAST. Each motif is represented by a specific color. (Color figure online)

4, 5, 10, and 12 were present in all TaGH3-III sub-family members. In addition, the same sub-family was observed with common motifs. The sequences, positions and markers of the conserved motifs (motif 1, 2, 5 and 12) in typical proteins of each sub-family of TaGH3 are shown in Supplemental Fig. S2.

Chromosomal locations of TaGH3 genes

The reference GFF3 files provided the chromosomal locations of TaGH3 genes. Based on this extracted physical location (Supplemental Table S3), A chromosome map of TaGH3 genes on 14 chromosomes was produced using MapInspect software (Fig. 3). The TaGH3s were distributed

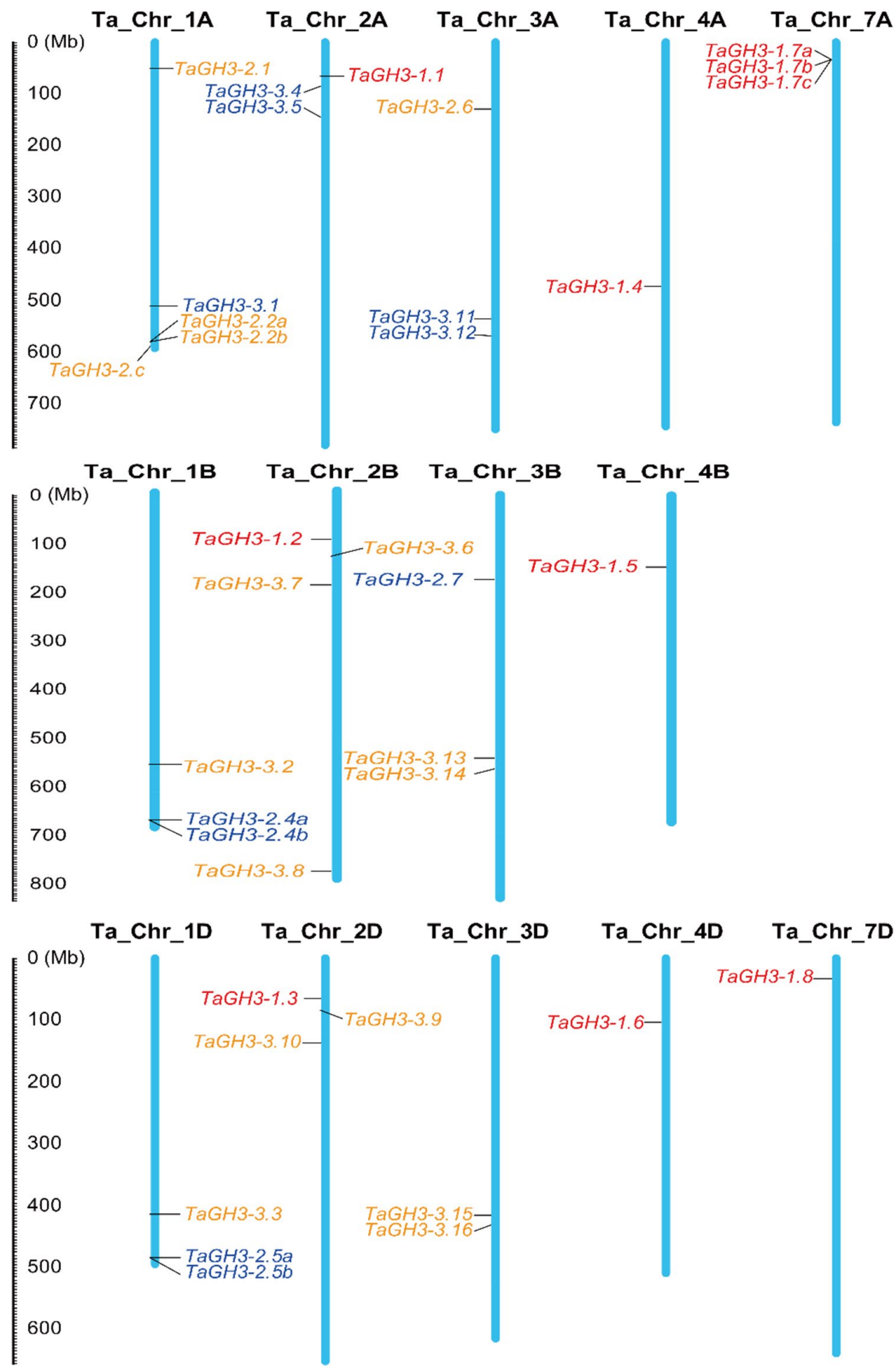


Fig. 3 Chromosome locations of the 36 *TaGH3s* in wheat. Different sub-groups of TaGH3s are represented in different colors: red, TaGH3-I; orange, TaGH3-II; blue, TaGH3-III. In addition. Chr, Chromosome. (Color figure online)

roughly evenly across the three sub-genomes (sub-genome A, 15; sub-genome B, 11; and sub-genome D, 10). Members from the same sub-group tended to be distributed at similar locations in the homoeologous chromosomes. However, the distribution range of genes on chromosomes varied from one homoeologous group to another. Only TaGH3 III members were distributed on homoeologous chromosome groups 4 and 7. The density of TaGH3 genes was highest on homoeologous group 1 (11 genes) at 30.56% of all TaGH3 genes (Fig. 3). We performed sequence homology analysis on 36 TaGH3 protein family members (Table S4). The distribution of TaGH3 genes on chromosome revealed a clustering pattern indicative of tandem replication.

The diploid species *Triticum urartu* is considered the source of the A genome in common wheat. *TuGH3* genes are present on all seven chromosomes. Chromosome 2 contained the highest number of *TuGH3* genes (4); and one member was assigned on each of chromosomes 1, 4, 5, 6, and 7. In *T. dicoccoides*, the tetraploid (AABB) progenitor of common wheat, 23 *TdGH3* genes were located on 10 chromosomes. Chromosomes 2A and 2B contained the highest number of TdGH3 genes (6); but no genes were on chromosomes 5B, 6A, 6B, and 7B. In *Aegilops tauschii*, the D-genome source for common wheat, 13 AetGH3 genes were unevenly distributed on seven chromosomes. The results showed that the distribution of *GH3* genes on chromosomes in the different species was somewhat irregular; but chromosomes in homoeologous group 2 consistently had more TaGH3 genes than the others (Figs. 3 and 4).

Identification of *Cis*-acting regulatory elements in the promoters of TaGH3 family genes

A *cis*-acting regulatory element is a specific motif that binds to an appropriate transcription factor to control gene transcription [66]. Several *cis*-elements involved in phytohormone response have been identified in plants. The distribution of different *cis*-acting elements in gene promoters may reflect differences in function and regulation. Identified *cis*-acting elements were divided into three major categories, namely those relating to biotic/abiotic stress, growth and phytohormone response. Some elements, such as ABRE [67], ARE [68], TGA-element [69], AuxRR-core [70], MYB [26], and Sp1 [71] were enriched in the light responsive *TaGH3* genes (Fig. 5). The G-box related to biotic/abiotic stress response was the most common motif in *TaGH3* gene promoters. Hormone-related motifs were also very common in the promoters. The auxin-responsive TGA-element and AuxRR-core are important regulatory elements in *TaGH3* genes. In addition, the TaGH3 gene promoter regions also had *cis*-acting elements that are involved in salicylic acid activity (TCA-element). The abscisic acid (ABA) responsive *cis*-acting element (ABRE) was also identified in most *TaGH3* genes [72]. The very common canonical AuxRR-core and TGA-element were found in promoters of 12 TaGH3 genes (*TaGH3-1.1*, *TaGH3-1.3*, *TaGH3-2.4a*, *TaGH3-2.5b*, *TaGH3-2.7*, *TaGH3-3.1*, *TaGH3-3.2, TaGH3-3.3*, *TaGH3-3.5*, *TaGH3-3.8, TaGH3-3.10* and *TaGH3-3.16*). Most TaGH3 genes, except *TaGH3-1.6* and *TaGH3-1.7a/b/c* contained ABRE *cis*-elements. This suggests that the expression of TaGH3 genes is regulated by stress-related hormones. Some growth and development-related *cis*-elements, such as CAAT-box, CAT-box and A-box, were present in the promoters of some *TaGH3* genes (Fig. 5). The numbers of stress related *cis*-elements in the upstream 1.5 Kb regions of TaGH3 genes are listed in Table S5.

Functional categorization of TaGH3 genes based on GO annotation

Multiple databases, including GO (Gene Ontology), were used for functional annotation of *TaGH3* genes [73, 74]. In the study, we conducted GO analysis on genes family of *TaGH3*, which helped us to better understand the genes function of these *TaGH3s*. GO terms related to: (1) Biological Processes (BP), including response to auxin (GO: 0009733), response to light stimulus (GO: 000416), induced systemic resistance, jasmonic acid mediated signaling (GO: 0009864), etc.;(2) Cellular Components (CC), including plastids (GO: 0009536), chloroplasts (GO: 0009507), vacuoles (GO: 0005773), chloroplast envelopes (GO: 0009941), and cytoplasmic membrane-bounded vesicles (GO: 0016023); (3) Molecular Function (MF), including indole-3-acetic acid amido synthetase activity (GO: 0010279), ligase activity (GO: 0016874), adenylyltransferase activity (GO: 0070566), etc. were specifically enriched (Fig. 6, Supplemental Table S6). GO analysis indicates that all members of the TaGH3-I sub-family are involved in two biological processes, response to light stimulus (GO: 0009416) and response to auxin (GO: 0009733). Among them, TaGH3-1.4, TaGH3-1.5 and TaGH3-1.6 also participated in salicylic acid mediated signaling pathway (GO: 0009863) and benzoate metabolic process (GO: 0018874). Interestingly, we found that *TaGH3-1.7* and *TaGH3-1.8* are involved in the biological processes of defense response to bacterium (GO: 0042742). In the TaGH3-II subfamily, except TaGH3-2.1 and TaGH3-2.3, the remaining eight members are involved in the biological processes of jasmonic acid metabolic process (GO: 0009694), photomorphogenesis (GO: 0009640) and response to ozone (GO: 0010193). At the same time, these eight TaGH3-II subfamily members also have jasmonate-amino synthetase activity (GO: 0080123), and we also found that TaGH3-2.1 has indole-3-acetic acid amido synthetase activity (GO: 0010279), and TaGH3-2.3 has ligase activity (GO: 0016874). Similarly, we found in the TaGH3-III subfamily that, like the I subfamily,

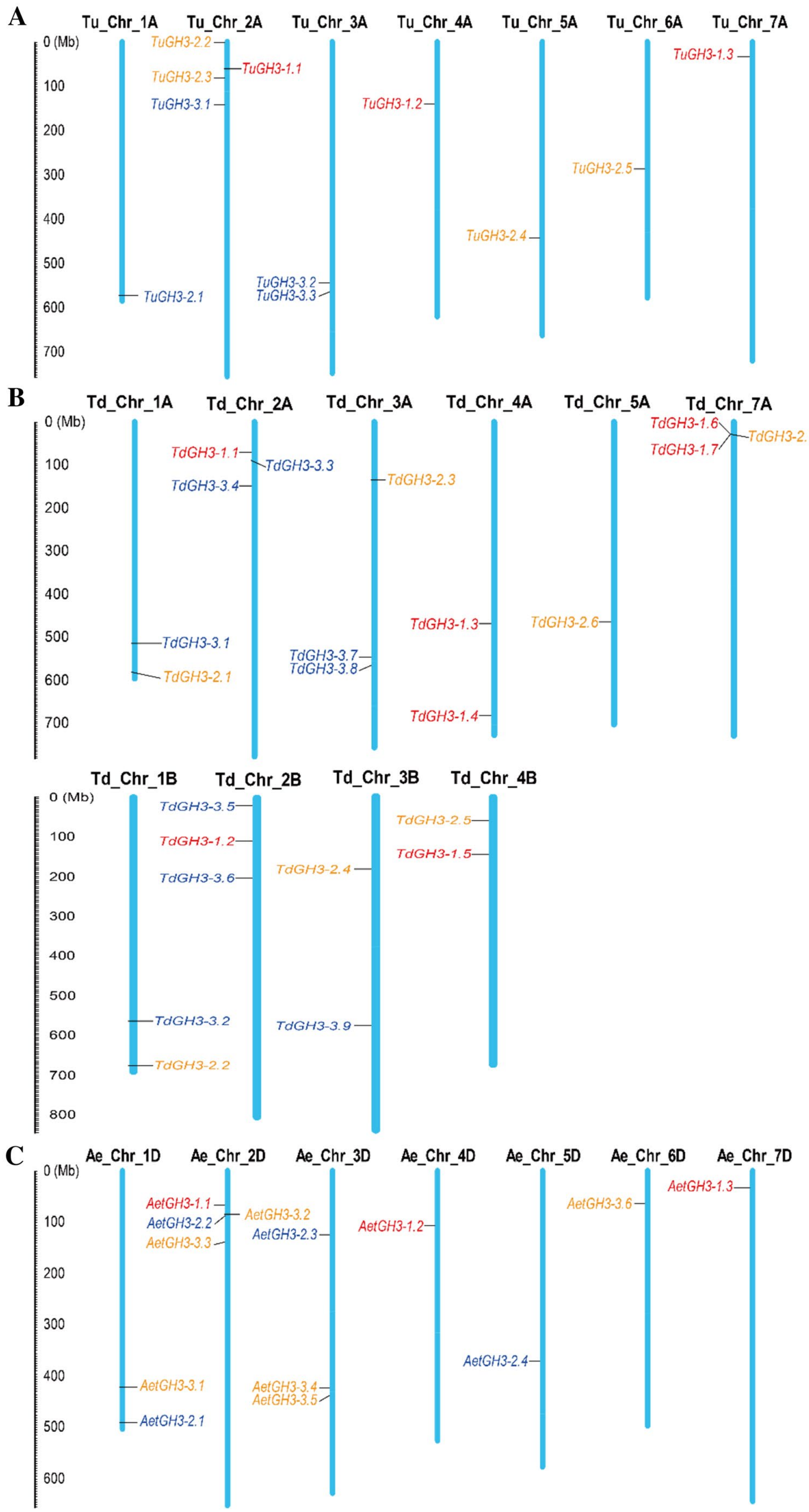


Fig. 4 Chromosomal locations of *TuGH3s*, *TdGH3s*, and *AetGH3s*. Chromosomal locations of **a** *TuGH3* genes; **b** *TdGH3* genes; **c** *AetGH3* genes. Different groups of *GH3s* are represented in different colors: red, TaGH3-I; orange, TaGH3-II; blue, TaGH3-III. Chr, Chromosome. (Color figure online)

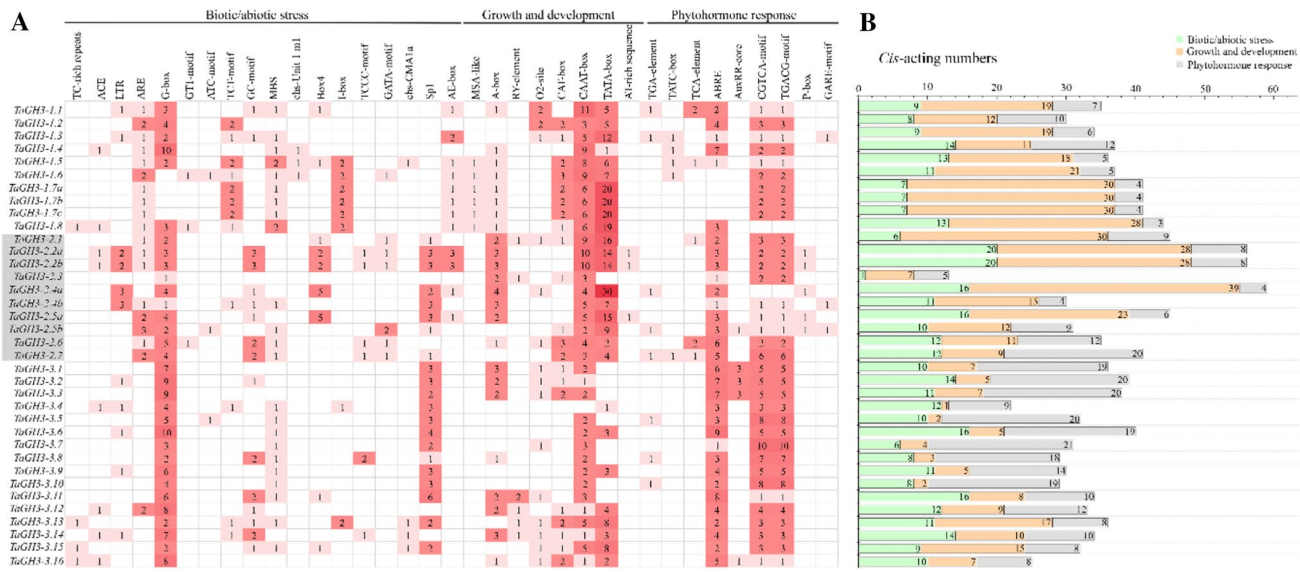


Fig. 5 Analysis of *cis*-acting element numbers in *TaGH3* genes. **a** Different colors and numbers on the grid indicate numbers of different promoter elements in each *TaGH3* gene. **b** Different colors on the histograms represent the sum of the *cis*-acting elements in each category. (Color figure online)

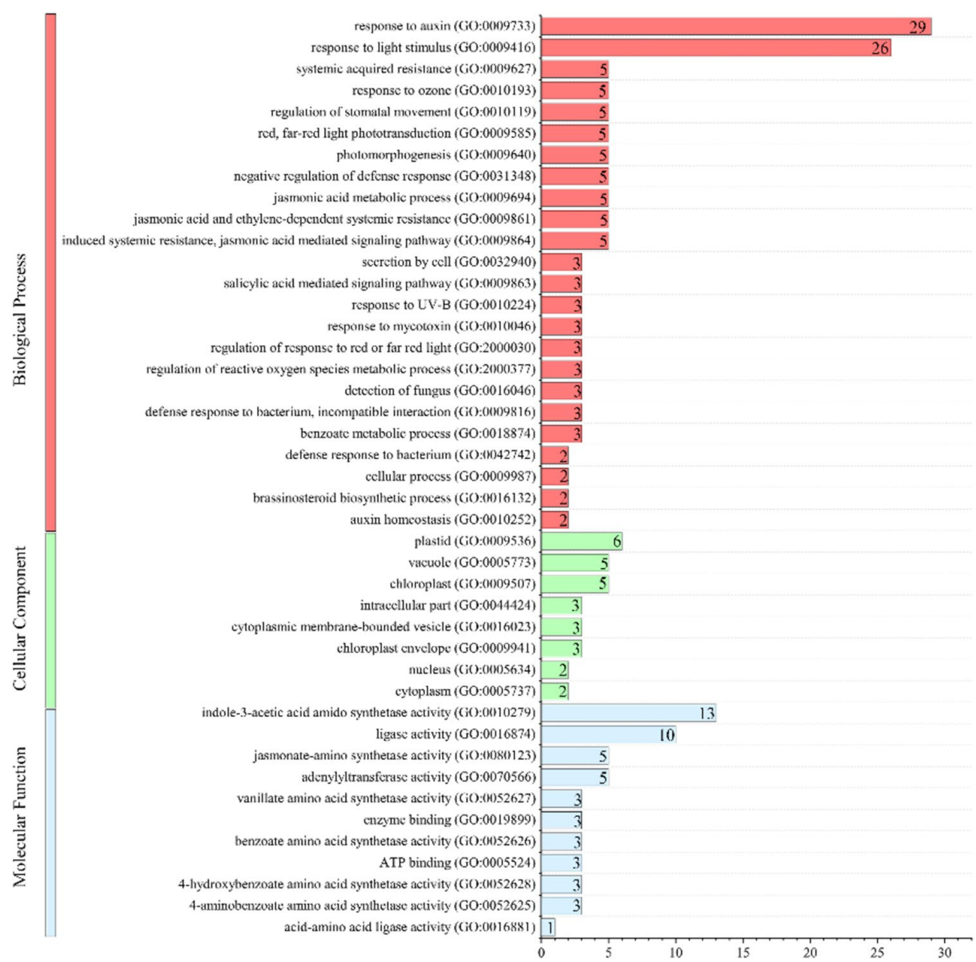


Fig. 6 Functional annotation analysis of TaGH3 genes. Gene Ontology (GO) classification based on *TaGH3* gene annotation. GO terms are summarized in three main categories

all members participate in response to light stimulus (GO: 0009416) and response to auxin (GO: 0009733). But we found that TaGH3-3.11, TaGH3-3.13, and TaGH3-3.15 are also involved in the special biological processes of secretion by cell (GO: 0032940). Overall, GO enrichment analysis showed that most TaGH3s were annotated under GO terms 'response to auxin' (GO: 0009733) and 'response to light stimulus' (GO: 0009416). These results are consistent with previous studies, suggesting that plant resistance is a complex regulatory network involving signaling, material and energy metabolism, and defense processes.

Homologous gene pairs and synteny analysis

In order to better understand the evolutionary constraints of the GH3 gene family, Ka/Ks analyses and replication times were determined for collinear sections. The ratio of non-synonymous substitution rate to the synonymous mutation rate (Ka/Ks) is usually used to estimate the evolutionary timing of a gene to test the environmental selection force experienced by the coding sequence [75]. Generally, a Ka/Ks ratio > 1 indicates accelerated evolution with positive selection, a ratio equal to 1 indicates neutral selection, and a ratio < 1 indicates negative or purifying selection. The Ka/Ks ratios for 42 pairs of replicated genes were < 1 (Table 2), indicating that the GH3 gene family mainly underwent purifying selection. However, the Ka/Ks ratio for TaGH3-1.7a and TaGH3-1.8 was 1.030, indicating that that pair of genes had undergone neutral evolution. In addition, we estimated the time at which a segment replication event corresponding to the GH3 gene occurred. Based on the molecular clock theory, the Ks value can be used to estimate the time of occurrence of segment repetitive events containing repetitive genes [76]. The results indicated that 43 copies of the *GH3* genes corresponding to the replication of the replication occurred approximately 0.31 to 9.35 million years ago (Table 2).

The syntenic analysis was expanded to the progenitor species for comparison with common wheat. The numbers of *GH3* genes identified in *T. urartu*, *T. dicoccoides* and *Ae. tauschii* gene family were 11, 23, and 13, respectively. The phylogenetic tree for all were established using the Mega 7.0 Maximum Likelihood (ML) method (Fig. 7a). The GH3 proteins can be divided into three broad categories (GH3-I, GH3-II, and GH3-III), *T. aestivum* (10, 10, and 16), *T. urartu* (3, 5, and 3), *T. dicoccoides* (7, 7, and 9), and *Ae. tauschii* (3, 4, and 6). Identical orthologues were found in the corresponding genomic regions among the A, B, D sub-genomes of wheat and the progenitor species (Fig. 7b, Table S7). Considering the fact that duplicated TaGH3 gene pairs were under negative selection force, it is apparent that polyploidization was the main reason for the higher number of TaGH3 family genes.

In silico expression pattern profiling of GH3 genes in wheat

Auxin-related transcriptional regulation is an essential process required for plants to survive and adapt to adverse environmental challenges [77]. We performed a comprehensive transcriptome analysis to estimate the expression patterns of each *TaGH3* gene in biotic stress, abiotic stress, and growth and development. The original transcriptome data was downloaded from the NCBI database and used to mine the expression profiling of 36 *TaGH3* genes (Table S8). Expression levels represented by FPKM were used to generate a heatmap (Figs. 8, 9).

In the transcriptome data analyses, the expression levels of genes in different groups were significantly different. In general, under various of biotic stresses, abiotic stresses, and growth and development, the expression levels of TaGH3-II sub-family genes were generally relatively good. The expression level of TaGH3-III subfamily genes were relatively low, which only had small expression levels under certain treatments. But under biological stress, especially under *Fusarium graminearum*, the expression level was relatively high. Members of the TaGH3-I sub-family, except *TaGH3-1.8*, the remaining genes were hardly expressed under various conditions.

As shown in Fig. 8 most *TaGH3* genes were expressed at most developmental stages, suggesting crucial roles in plant growth and development. Of course, we could find that some genes (*TaGH3-1.5/1.6/1.7b*, *TaGH3-2.2b/2.3*, and *TaGH3-3.4/3.6/3.9*) were not expressed throughout the growth and development process. More often, some *TaGH3* genes were expressed were highly under specific conditions; for example, expression of *TaGH3-1.7*, *TaGH3-1.8* and *TaGH3-2.2* was up-regulated at the seeding stage, *TaGH3-1.8*, *TaGH3-2.2a/2.4b*, and *TaGH3-3.1/3.3/3.13*genes were highly expressed at full booting, and *TaGH3-2.5a* and *TaGH3-3.1* were specifically induced in anthesis-associated processes. *TaGH3-1.8*, *TaGH3-3.5*, and *TaGH3-3.10* showed highest expression levels at the milk grain stage (Fig. 8). *TaGH3-1.7*, *TaGH3-1.8*, *TaGH3-2.2a*, *TaGH3-2.4b*, *TaGH3-2.5a*, *TaGH3-2.5b*, *TaGH3-2.6*, *TaGH3-2.7*, *TaGH3-3.3*, *TaGH3-3.5*, *TaGH3-3.7*, *TaGH3-3.11*, *TaGH3-3.13* and *TaGH3-3.15* were highly expressed in multiple tissues and were likely involved in regulation of growth and development.

Expression of *TaGH3* genes under abiotic stress is shown in Fig. 9. Under osmotic stress simulated by PEG6000, expression of genes *TaGH3-1.8*, *TaGH3-2.2a*, *TaGH3-2.4b*, *TaGH3-2.5a*, *TaGH3-3.7*, *TaGH3-3.13* and *TaGH3-3.15* was generally increased (fold change > 3) but different groups of genes showed different. Expression of *TaGH3-1.7a* and *TaGH3-3.15* in cv. Gemmiza_10 increased with prolonged treatment time, whereas expression in in cv. Giza_168 decreased with time. Under high temperature stress genes *TaGH3-1.8*, *TaGH3-2.5b*, *TaGH3-3.10* and *TaGH3-3.15* were highly

Table 2 Ka/Ks analysis and estimated divergence time for the duplicated GH3 gene pairs

Duplicated GH3 gene pairs	Ka	Ks	Ka/Ks	Duplication time/ (Million years)	Duplicate type
TaGH3-1.1 & *TaGH3-1.2*	0.005	0.059	0.086	0.45	Segmental
TaGH3-1.1 & *TaGH3-1.3*	0.011	0.06	0.183	0.46	Segmental
TaGH3-1.2 & *TaGH3-1.3*	0.008	0.041	0.191	0.31	Segmental
TaGH3-1.4 & *TaGH3-1.5*	0.024	0.071	0.344	0.54	Segmental
TaGH3-1.4 & *TaGH3-1.6*	0.02	0.061	0.326	0.46	Segmental
TaGH3-1.5 & *TaGH3-1.6*	0.014	0.057	0.25	0.44	Segmental
TaGH3-1.7a & *TaGH3-1.8*	1.008	0.978	1.03	7.53	Segmental
TaGH3-2.1 & *TaGH3-2.3*	0.029	0.042	0.684	0.32	Segmental
TaGH3-2.2a & *TaGH3-2.4a*	0.021	0.109	0.191	0.84	Segmental
TaGH3-2.2a & *TaGH3-2.5a*	0.019	0.142	0.137	1.09	Segmental
TaGH3-2.4a & *TaGH3-2.5a*	0.009	0.101	0.089	0.77	Segmental
TaGH3-2.6 & *TaGH3-2.7*	0.008	0.191	0.041	1.47	Segmental
TaGH3-3.1 & *TaGH3-3.2*	0.963	1.092	0.882	8.39	Segmental
TaGH3-3.1 & *TaGH3-3.3*	0.966	1.083	0.891	8.33	Segmental
TaGH3-3.11 & *TaGH3-3.13*	0.007	0.236	0.032	1.81	Segmental
TaGH3-3.11 & *TaGH3-3.15*	0.005	0.108	0.045	0.82	Segmental
TaGH3-3.12 & *TaGH3-3.14*	0.009	0.192	0.045	1.48	Segmental
TaGH3-3.12 & *TaGH3-3.16*	0.004	0.23	0.017	1.77	Segmental
TaGH3-3.13 & *TaGH3-3.15*	0.007	0.234	0.029	1.8	Segmental
TaGH3-3.14 & *TaGH3-3.16*	0.007	0.225	0.029	1.73	Segmental
TaGH3-3.2 & *TaGH3-3.3*	0.004	0.242	0.015	1.86	Segmental
TaGH3-3.5 & *TaGH3-3.10*	0.002	0.201	0.01	1.54	Segmental
TaGH3-3.5 & *TaGH3-3.7*	0.002	0.284	0.007	2.18	Segmental
TaGH3-3.6 & *TaGH3-3.7*	0.103	1.216	0.085	9.35	Tandem
TaGH3-3.7 & *TaGH3-3.10*	0.002	0.234	0.01	1.8	Segmental
OsGH3-1 & *TaGH3-2.6*	0.088	0.253	0.346	1.95	Segmental
OsGH3-1 & *TaGH3-2.7*	0.095	0.278	0.342	2.14	Segmental
OsGH3-5 & *TaGH3-3.1*	0.114	0.466	0.245	3.58	Segmental
OsGH3-5 & *TaGH3-3.2*	0.114	0.467	0.244	3.59	Segmental
OsGH3-5 & *TaGH3-3.3*	0.116	0.47	0.247	3.61	Segmental
ZmGH3-1 & *TaGH3-3.5*	0.097	0.357	0.273	2.74	Segmental
ZmGH3-7 & *TaGH3-3.3*	0.346	0.594	0.583	4.57	Segmental
ZmGH3-7 & *TaGH3-3.5*	0.091	0.346	0.263	2.67	Segmental
ZmGH3-7 & *TaGH3-3.7*	0.089	0.39	0.229	3	Segmental
ZmGH3-12 & *TaGH3-3.11*	0.07	0.329	0.212	2.53	Segmental
OsGH3-3 & *TaGH3-3.12*	0.063	0.279	0.227	2.14	Segmental
ZmGH3-11 & *TaGH3-3.12*	0.063	0.417	0.151	3.21	Segmental
ZmGH3-12 & *TaGH3-3.13*	0.07	0.365	0.192	2.81	Segmental
OsGH3-3 & *TaGH3-3.14*	0.061	0.3	0.202	2.3	Segmental
ZmGH3-11 & *TaGH3-3.14*	0.063	0.436	0.144	3.35	Segmental
ZmGH3-12 & *TaGH3-3.15*	0.071	0.325	0.217	2.5	Segmental
OsGH3-3 & *TaGH3-3.16*	0.062	0.272	0.227	2.09	Segmental
ZmGH3-11 & *TaGH3-3.16*	0.062	0.39	0.158	3	Segmental

expressed, and their expression levels at 6 h post treatment were higher than those at 1 h. Expression levels of *TaGH3-1.8*, *TaGH3-2.6*, *TaGH3-2.7* and *TaGH3-3.8* were down-regulated after increasing heat treatment compared to osmotic stress only after 1 h. After 2 weeks of low temperature treatment, expression levels of *TaGH3-1.8* and *TaGH3-2.2a* were significantly higher than the controls. With phosphorus starvation 3 genes *TaGH3-2.2a*, *TaGH3-3.5* and *TaGH3-3.7* were more highly expressed in roots compared to the controls (Fig. 9a).

During the biotic stress, Twenty-four *TaGH3* genes showed increased expression following infection with

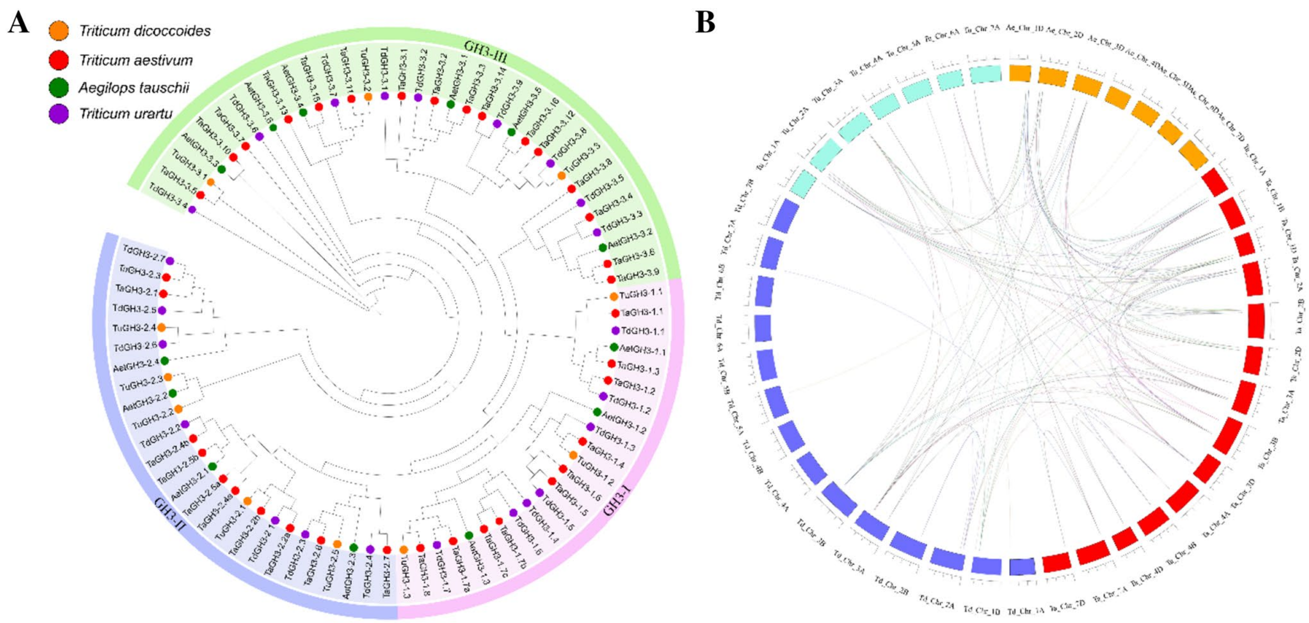


Fig. 7 Phylogeny of the GH3 gene family in wheat and its progenitors. **a** Phylogenetic relationship of *Triticum aestivum*, *T. urartu*, *T. dicoccoides*, and *Ae. tauschii*. Protein sequences were aligned using the ClustalW2 sequence alignment program and the phylogenetic tree constructed by software MEGA7 was used to create Maximum Likelihood (ML) under the LG model. The tree was constructed with 1000 bootstrap replications. **b** Orthologous relationships between wheat and its genomic progenitors. Numbers along each chromosome box indicate sequence lengths in megabases. Lines joining chromosomes represent the syntenic relationships. *T. aestivum*, *T. urartu*, *T. dicoccoides*, and *Ae. tauschii* chromosomes are shown in different colored boxes; Ta, red; Td, purple; Tu, cyan; Ae, orange. (Color figure online)

Fusarium graminearum (Fig. 9b). The heatmap showed that the *TaGH3s* could be divided into two groups. One group of 10 members were highly expressed in cv. Gemmiza_10 and Giza_168, and another group of 14 members had lower expression. Thirteen *TaGH3* genes (*TaGH3-1.1*, *TaGH3-1.2*, *TaGH3-1.3*, *TaGH3-1.8*, *TaGH3-2.2*, *TaGH3-2.4a*, *TaGH3-2.4b*, *TaGH3-2.5a*, *TaGH3-2.5b*, *TaGH3-2.6*, *TaGH3-2.7*, *TaGH3-3.3*, *TaGH3-3.5*, *TaGH3-3.7*, *TaGH3-3.8* and *TaGH3-3.10*) responded to infection by stripe rust, but differed in expression at different time points. For example, *TaGH3-2.2a* showed higher expression level on the first day of treatment, whereas *TaGH3-3.3* gene was significantly expressed on the ninth day post infection of cv. Vuka. At 14 days post inoculation (dpi) with *Zymoseptoria tritici*, expression of *TaGH3-1.8*, *TaGH3-2.4b*, *TaGH3-2.5b*, *TaGH3-2.6* and *TaGH3-2.7* genes was inhibited compared to the control. Expression of *TaGH3-1.8* increased from one dpi infection by powdery mildew (Fig. 9b).

qRT-PCR confirmed the response capability of TaGH3s to stress conditions

High salinity and drought are the major environmental stresses affecting growth and development of crop plants under natural conditions [78]. Treatment of wheat seedlings with 0.15 M/L NaCl, 0.18 M/L mannitol, and 20% PEG at the one-leaf stage caused many changes in seedling growth status. At 12 h post treatment leaves softened and turned yellow, and growth inhibition started. After 72 h, leaf moisture was lost, and the leaves became brittle (Fig. 10a).

Analysis of *TaGH3* gene expression provided insight into the roles of different tissues in different environmental stresses of wheat. We verified the responses of candidate proteins to salt and osmotic stresses and analyzed their functions in different plant tissues by qRT-PCR. Expression of 8 *TaGH3* genes was studied in these treatments (Fig. 10b and Table S9). Each treatment caused different expression patterns in wheat leaf tissues. Five of the genes (*TaGH3-1.8*, *TaGH2.4a*, *TaGH3-3.7*, *TaGH3-3.11* and *TaGH3-3.15*) showed high expression at 72 h post beginning salt and drought treatments. However, the expression level of *TaGH3-2.2a* was very low until 120 h of treatment, indicating that the gene would play an important role in the late growth of wheat leaves. All eight genes were up-regulated at different times in root tissue. Compared with the control, expression of *TaGH3-3.7* gene was up-regulated after 12 h of drought treatment. As the treatment time increased, the expression level gradually increased until 120 h. Down-regulation of *TaGH3-2.2a* expression was observed only in the

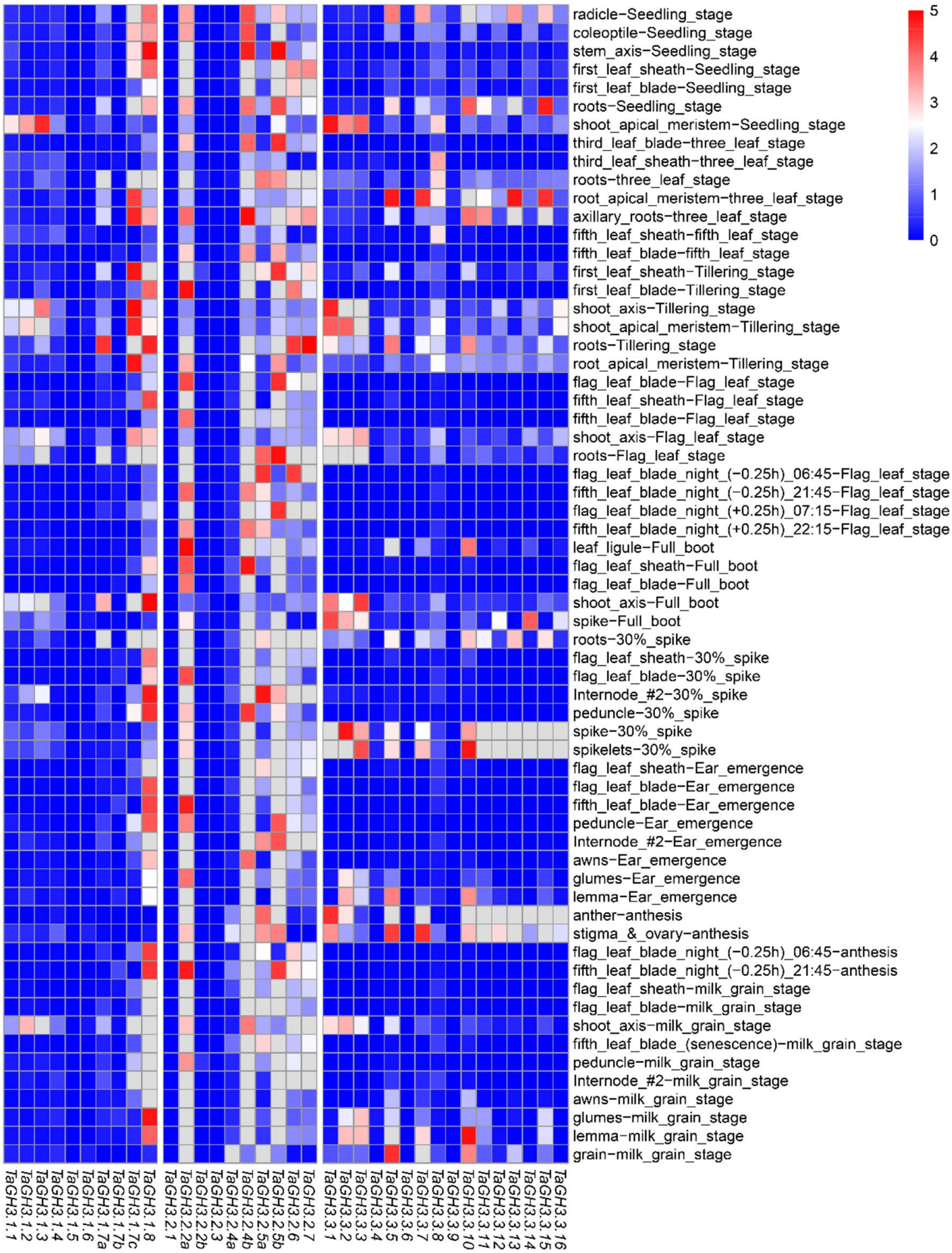
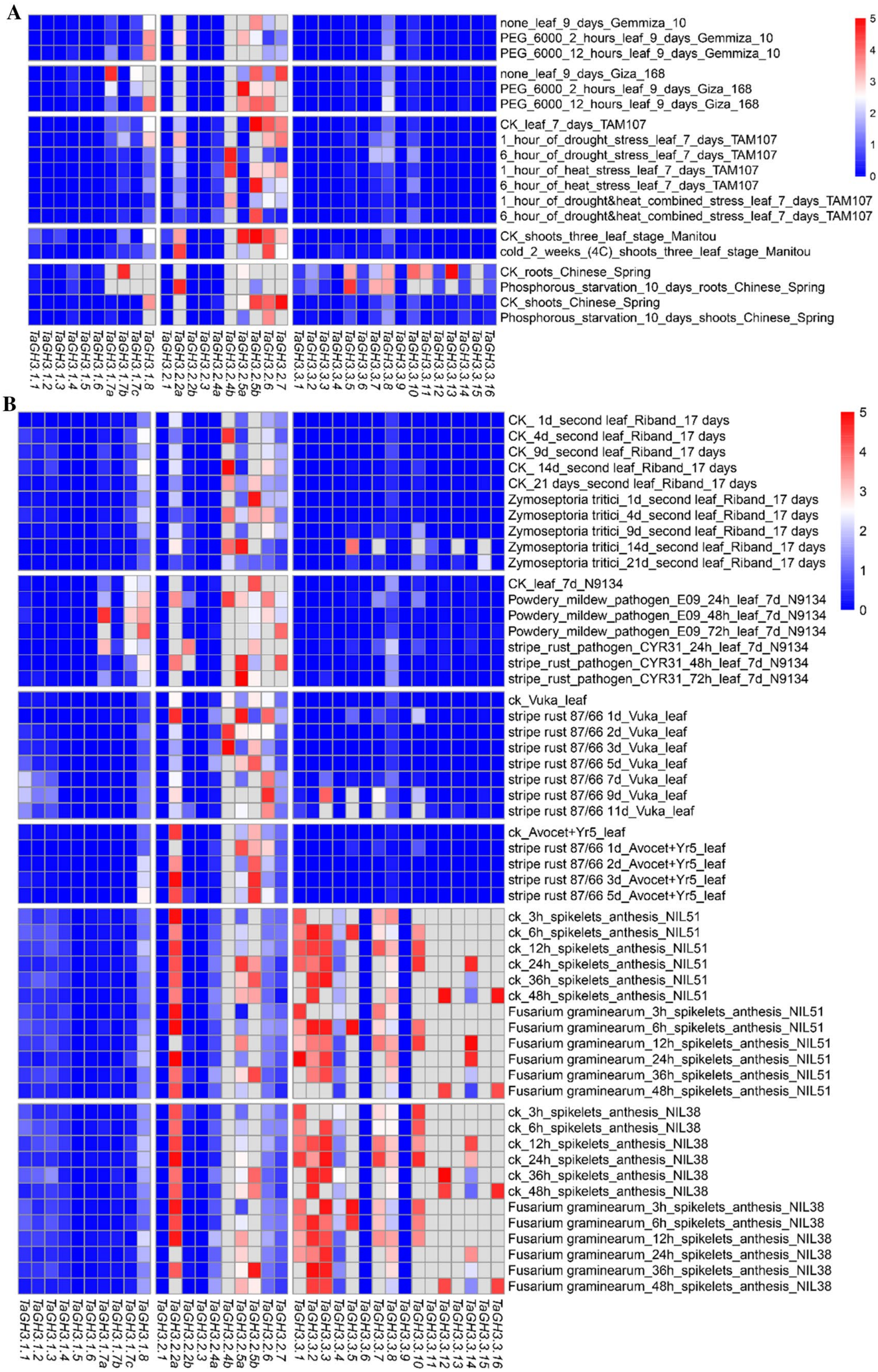


Fig. 8 Expression of *TaGH3* genes in various tissues from cv. Chinese Spring at different growth stages. Red color indicates increased expression levels; blue color indicates decreased expression levels. A fold change > 3 was considered significantly expressed. (Color figure online)

early stages of drought and salt stress treatment, whereas all other *TaGH3* genes were up-regulated. Overall, these genes were capable of responding to drought and salt stress, and were generally highly expressed under osmotic stress.


A
none_leaf_9_days_Gemmiza_10
PEG_6000_2_hours_leaf_9_days_Gemmiza_10
PEG_6000_12_hours_leaf_9_days_Gemmiza_10
none_leaf_9_days_Giza_168
PEG_6000_2_hours_leaf_9_days_Giza_168
PEG_6000_12_hours_leaf_9_days_Giza_168
CK_leaf_7_days_TAM107
1_hour_of_drought_stress_leaf_7_days_TAM107
6_hour_of_drought_stress_leaf_7_days_TAM107
1_hour_of_heat_stress_leaf_7_days_TAM107
6_hour_of_heat_stress_leaf_7_days_TAM107
1_hour_of_drought&heat_combined_stress_leaf_7_days_TAM107
6_hour_of_drought&heat_combined_stress_leaf_7_days_TAM107
CK_shoots_three_leaf_stage_Manitou
cold_2_weeks_(4C)_shoots_three_leaf_stage_Manitou
CK_roots_Chinese_Spring
Phosphorous_starvation_10_days_roots_Chinese_Spring
CK_shoots_Chinese_Spring
Phosphorous_starvation_10_days_shoots_Chinese_Spring
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TaGH3.1.1
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TaGH3.1.7a
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TaGH3.1.7c
TaGH3.1.8
TaGH3.2.1
TaGH3.2.2a
TaGH3.2.2b
TaGH3.2.3
TaGH3.2.4a
TaGH3.2.4b
TaGH3.2.5a
TaGH3.2.5b
TaGH3.2.6
TaGH3.2.7
TaGH3.3.1
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TaGH3.3.8
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TaGH3.3.10
TaGH3.3.11
TaGH3.3.12
TaGH3.3.13
TaGH3.3.14
TaGH3.3.15
TaGH3.3.16
B
CK_ 1d_second leaf_Riband_17 days
CK_4d_second leaf_Riband_17 days
CK_9d_second leaf_Riband_17 days
CK_ 14d_second leaf_Riband_17 days
CK_21 days_second leaf_Riband_17 days
Zymoseptoria tritici_1d_second leaf_Riband_17 days
Zymoseptoria tritici_4d_second leaf_Riband_17 days
Zymoseptoria tritici_9d_second leaf_Riband_17 days
Zymoseptoria tritici_14d_second leaf_Riband_17 days
Zymoseptoria tritici_21d_second leaf_Riband_17 days
CK_leaf_7d_N9134
Powdery_mildew_pathogen_E09_24h_leaf_7d_N9134
Powdery_mildew_pathogen_E09_48h_leaf_7d_N9134
Powdery_mildew_pathogen_E09_72h_leaf_7d_N9134
stripe_rust_pathogen_CYR31_24h_leaf_7d_N9134
stripe_rust_pathogen_CYR31_48h_leaf_7d_N9134
stripe_rust_pathogen_CYR31_72h_leaf_7d_N9134
ck_Vuka_leaf
stripe rust 87/66 1d_Vuka_leaf
stripe rust 87/66 2d_Vuka_leaf
stripe rust 87/66 3d_Vuka_leaf
stripe rust 87/66 5d_Vuka_leaf
stripe rust 87/66 7d_Vuka_leaf
stripe rust 87/66 9d_Vuka_leaf
stripe rust 87/66 11d_Vuka_leaf
ck_Avocet+Yr5_leaf
stripe rust 87/66 1d_Avocet+Yr5_leaf
stripe rust 87/66 2d_Avocet+Yr5_leaf
stripe rust 87/66 3d_Avocet+Yr5_leaf
stripe rust 87/66 5d_Avocet+Yr5_leaf
ck_3h_spikelets_anthesis_NIL51
ck_6h_spikelets_anthesis_NIL51
ck_12h_spikelets_anthesis_NIL51
ck_24h_spikelets_anthesis_NIL51
ck_36h_spikelets_anthesis_NIL51
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Fusarium graminearum_48h_spikelets_anthesis_NIL51
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ck_24h_spikelets_anthesis_NIL38
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Fusarium graminearum_3h_spikelets_anthesis_NIL38
Fusarium graminearum_6h_spikelets_anthesis_NIL38
Fusarium graminearum_12h_spikelets_anthesis_NIL38
Fusarium graminearum_24h_spikelets_anthesis_NIL38
Fusarium graminearum_36h_spikelets_anthesis_NIL38
Fusarium graminearum_48h_spikelets_anthesis_NIL38
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TaGH3.1.1
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TaGH3.1.7a
TaGH3.1.7b
TaGH3.1.7c
TaGH3.1.8
TaGH3.2.1
TaGH3.2.2a
TaGH3.2.2b
TaGH3.2.3
TaGH3.2.4a
TaGH3.2.4b
TaGH3.2.5a
TaGH3.2.5b
TaGH3.2.6
TaGH3.2.7
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TaGH3.3.16

◂**Fig. 9** Transcriptome analyses of 36 *TaGH3s* under multiple conditions. **a** Expression patterns of *TaGH3* genes in wheat plants subjected to different abiotic stress conditions, including drought stress (cv. Gemmiza 10; Giza 168; TAM 107), heat stress (cv.: TAM 107), drought and heat combined stress (cv. TAM 107), cold stress (cv. Manitou), and phosphorous starvation (cv. Chinese Spring). **b** Heat map of expression profiles for *TaGH3* genes under different stresses, including leaves of plants with with *Septoria tritici* blotch infection (cv. Riband), powdery mildew infection leaf (cv. Riband), and stripe rust infection (cv. Vuka and an Avocet introgression line containing resistance gene *Yr5*), and plants with Fusarium head blight infection at post-anthesis (cv. Remus) (SBA numbers: PRJEB8798, PRJNA243835, PRJNA243835, PRJEB12497 and PRJEB12358). Red color indicates increased expression levels; blue color indicates decreased expression levels. A fold change > 3 was considered significantly expressed. (Color figure online)

Homology modelling of TaGH3s

All 36 TaGH3 member proteins were three-dimensionally modelled using Phyre2 server in intensive mode (Supplemental Fig. S2). Predicted models were based on the following templates to heuristically maximize alignment coverage, percentage identity, and confidence score for the tested sequences. Template 5eci.2.A, 5ecn.1.A, 5ecl.2.A and 5ech.1.A (from *Arabidopsis thaliana*) was used in TaGH3-I subfamily modelling [79]; template 5kod.1.A (from *Arabidopsis thaliana*) was used for the TaGH3-II subfamily genes TaGH3-2.1 and TaGH3-2.3 [80]; and template 6e1q.1.A (from *Arabidopsis thaliana*) was used for TaGH3-2.2a/b, TaGH3-2.4a/b, TaGH3-2.5a/b, TaGH3-2.6, and TaGH3-2.7 [80]. Template 5kod.1.A (from *Arabidopsis thaliana*) were used for the TaGH3-III subfamily genes TaGH3-3.1, TaGH3-3.2, TaGH3-3.3, TaGH3-3.12, TaGH3-3.14 and TaGH3-3.16; template 4b2g.1.A (from *Arabidopsis thaliana*) was used for modeling the remaining TaGH3 subfamily member proteins [79]. However, it was apparent that to construct more reliable, native-like models the more experimentally solved structures are required from Gretchen hagen3 family proteins in particular from plant GH3s.

In all TaGH3 gene family members α-helices mainly determined the secondary structures of modelled wheat proteins with 33–46% whereas β-strands distributed with the 8–23%. Moreover, to determine the similarity or divergence of generated models, structures were superimposed to calculate the percentages of structure coverage. The superimposed TaGH3-I subfamily models showed 90–96% structural coverage; the superimposed TaGH3-I subfamily models were mainly demonstrated a 74–94% structural coverage. And the superimposed TaGH3-I subfamily models were mainly demonstrated a 74–100% structural coverage. TaGH3-3.9 structural coverage was 100% and every subfamily had a high coverage rate. Taken together, it has been implicated that GH3s from each genome donor either may have been ancestrally similar to each other or originally divergent GH3s could have been stabilized during long domestication process resulting in changes on protein structures thereby on protein functions.

Discussion

Auxin is an important plant hormone. Studies have shown that auxin is not only related to the light-morphological composition, apical dominance and root development, but also forms a complex signal transduction network with other hormones [81, 82]. GH3 (Gretchen Hagen 3) genes are a family of auxin-driven early response genes that encode a group of proteins that directly affect the homeostasis of other plant hormones [83]. However, the specific function(s) of most GH3 genes remains unknown. Prediction of GH3 genes genome-wide became possible with completion of genome sequencing. Wheat is prone to a number of abiotic stresses that prevent the achievement of yield potential [37]. The plant *GH3* genes plays important role in enhancing plant stress resistance and optimizing plant growth and metabolism, which makes the study of the entire wheat GH3 (TaGH3) gene family important [84]. Systematically analyzed the TaGH3 family genes, predicted the TaGH3 gene characteristics, chromosomal locations, sub-cellular localization, gene duplication events, *cis*-elements, and generated a phylogenetic tree and made a conserved domain analysis. And further analysis of the expression pattern of TaGH3 family genes under various abiotic stresses, which gives us the opportunity to identify candidate genes that play an important role in abiotic stress.

Thirty six *GH3* genes were identified in the wheat genome based on a search of the genome database of wheat and a comprehensive understanding of conserved domains. GH3 genes have been identified in many plants; the number varies widely among species. The number in wheat was similar to *Gossypium raimondii* (35 members) but many more than *Arabidopsis*, maize, and rice [85]. Therefore, we speculate that in the evolution from monocots to dicots the common ancestor of GH3s may evolve independently between different species. According to the general trend, GH3 protein first appeared in moss, and more *GH3* genes were discovered as plants continued to evolve. As research progresses, more and more researchers believe that repetitive sequences play an important role in the entire genome [86]. Gene replication events are major drivers of the evolution of genomes and genetic systems [87, 88]. Wheat has a complex evolutionary history, with two major polyploidisation events. The first is the older genome replication, which occurred before the genetic grouping of the herb (50–70 million years ago), which eventually formed an ancient doubling event [89]. The second time was that the traceability of common wheat originated from 300,000 years ago by the sub-genome

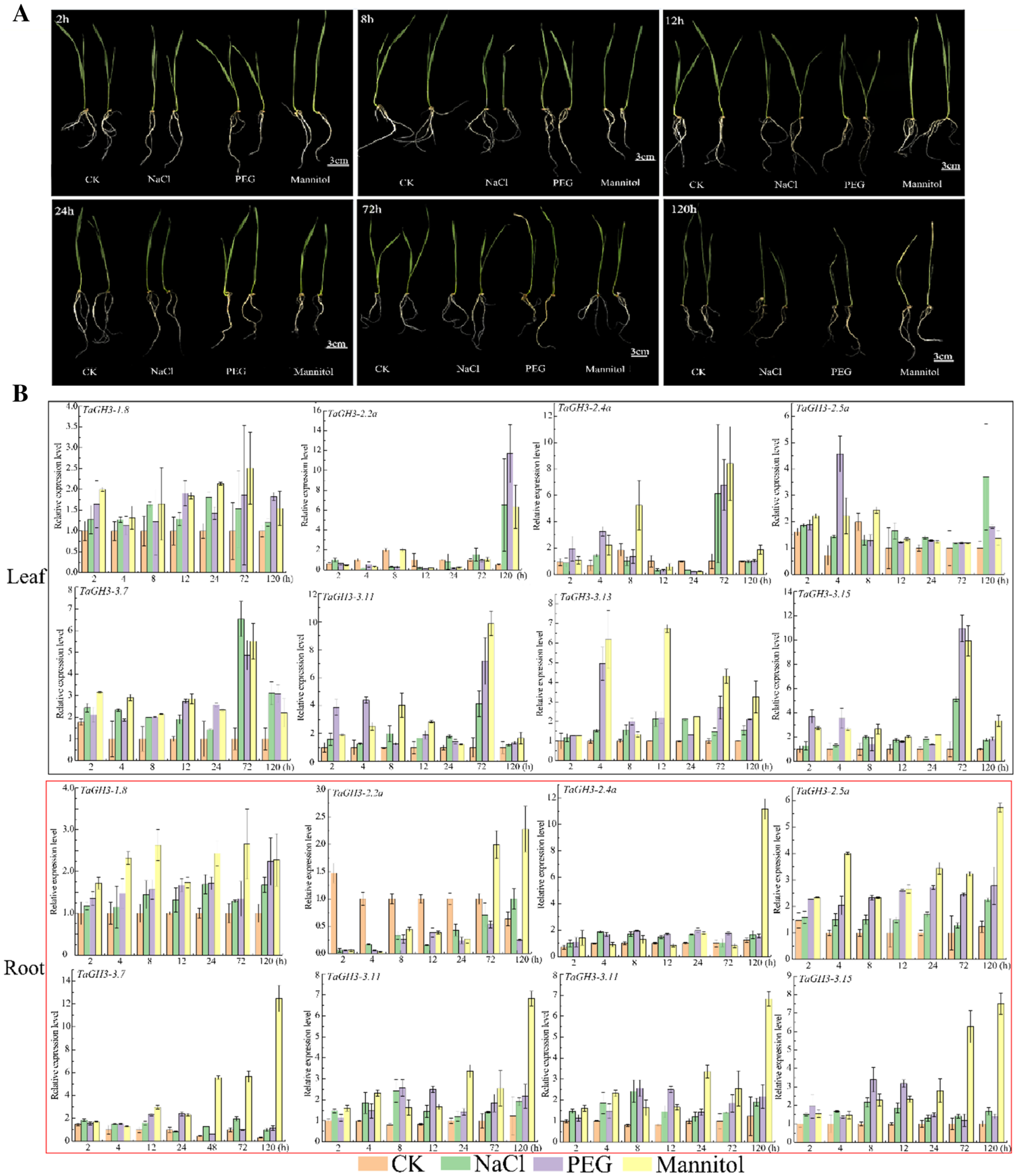

Fig. 10 The qRT-PCR results. **a** The apparent characteristics of wheat tissue culture at different time stages were treated by NaCl, PEG and Manntiol; **b** qRT-PCR analyses of *TaGH3* genes in in leaves and roots of plants stressed by NaCl, PEG and mannitol for indicated time periods shown on the x-axis. Expression levels are on the y-axis. Different tissues are displayed in different colored boxes, with black squares representing leaf tissue and red squares representing root tissue. Standard deviations are shown with error bars. The expression levels of *TaGH3* genes were plotted using Origin software. (Color figure online)

progenitor *Triticum urartu* (A sub-genome, controlling yield and disease resistance) and the hybridization of the *Aegilops speltoides* to form tetraploid wheat (*T. dicoccoides*, A and B sub-genome); then about 8000 years ago, the tetraploid wheat was again *Aegilops tauschii* (D sub-genome, control quality and adaptability) produces hexaploid wheat (*T. aestivum*, A, B, and D sub-genome) by natural hybridization [90]. A comparison of GH3 genes of hexaploid wheat and its diploid (*T. urartu* and *Aegilops tauschii*) and tetraploid (*T. dicoccoides*) progenitors showed that some genes were deleted during polyploidization (Fig. 7, Supplemental Table S7). This was likely caused by the allohexaploid genome of wheat.

We identified proteins from the wheat GH3 gene family and found that all GH3 proteins have a typical GH3 domain. Multiple sequence alignments on the wheat GH3 family genes indicated that all the genes had high amino acid sequence similarity (Fig. S1). Analysis of protein sequences encoded by all 36 *TaGH3* genes showed the presence of many conserved domains, consistent with the classification of GH3 family genes (Fig. 2, Supplemental Fig. S2, Table S2). Gene structure analysis revealed 1 to 5 introns, except for TaGH3-2.3 and TaGH3-3.8 with none (Fig. 2b). Despite the exon–intron variation the coding sequences was quite conserved. A large number of conserved domains are contained in most of TaGH3, except that TaGH3-2.1/2.3 and TaGH3-3.8/3.9 contain only 3 to 5 conserved domains. The conserved domain of TaGH3 protein may play an important physiological role in plant life activities. The function of protein is closely related to its localization in the cell. The newly synthesized protein must be transported to a specific organelle (ie, subcellular) in order to perform its function correctly. Therefore, predicting the subcellular localization of a protein has great significance in determining the function of an unknown protein, understanding protein interactions, and then understanding various biological processes and studying physiological mechanisms. We use Plant-mPLoc software to make subcellular localization predictions of wheat's GH3 protein. The prediction results indicate that the wheat GH3 protein is mainly localized in the chloroplast, and a small number of genes are localized in the nucleus and mitochondria (Table 1). The chloroplast is an important place for photosynthesis. Previous studies have shown that the protein encoded by the *FIN219* (*AtGH3-11*) gene of Arabidopsis thaliana is a component of the phyA pathway and is involved in the inhibition of the gene *COP1* by phyA-mediated far red light. *COP1* is an inhibitor of plant photomorphogenesis, so the protein encoded by *FIN219* is related to plant photomorphogenesis [91]. In addition, the expression of *OsJar1* gene homologous to *FIN219* in rice is also controlled by phyA, and it can up-regulate expression in response to methyl jasmonate treatment [78]. This indicates that the OsJar1 gene can regulate both the light signal and the jasmonate signal pathway during the photomorphogenesis of rice. These studies indicate that GH3 protein can respond to light signals, and its function may be related to its localization on chloroplasts. It also indicates the diversity of GH3 protein functions.

Systematic classification of TaGH3 proteins should help to understand the entire gene family. Staswick [2005] classified *Arabidopsis* GH3 proteins into three categories based on sequence similarity and specificity of the substrates with which they interacted. We generated phylogenetic trees based on *Arabidopsis*, rice, maize, and wheat and identified three groups (Fig. 1). The wheat and rice trees were very similar indicating that the functions of the *TaGH3* genes could be inferred from rice. In addition, phylogenetic analysis with other species of GH3 found that most of the wheat GH3 could find homologous sequences in *Arabidopsis*, maize or rice, suggesting that TaGH3s probably have the same functions as GH3s in other plant species. It is well known that the number of genes and the location of genes in most *Gramineae* species are relatively conservative. Therefore, the identification of the *GH3* genes from wheat or its orthologs from other species will provide more evidence for the identification of more orthologous genes from other *Gramineae* species.

Protein is the basic unit of life activity, and its structure determines its function. The study of protein structure is helpful for the study of its function. Analyzing the spatial structure of proteins is of great significance for understanding the functions of proteins, the execution of functions, and the interaction between biological macromolecules. Protein homology modeling is of great significance for the prediction of the function of advanced structural proteins. The prediction results showed that there are four main models (5eci.2.A, 5ecn.1.A, 5ecl.2.A, and 5ech.1.A) of protein structure predicted by the TaGH3-I sub-family, and the functions of these proteins are described as Jasmonic acid-amido synthetase (JAR1). There are two models (5kod.1.A and 6e1q.1.A) for predicting the structure of TaGH3-II sub-family member proteins, and their functions are described as Indole-3-acetic acid-amido synthetase GH3.5 and AtGH3.15 acyl acid amido synthetase. Similarly, we found two protein structure models (5kod.1.A and 4b2g.1.A) in the TaGH3-III sub-family, which correspond to the functions of Indole-3-acetic acid-amido synthetase GH3.5 and GH3-1 Auxin conjugating enzyme. These results suggest that the functions of wheat GH3 family proteins are mainly concentrated in Jasmonic acid-amido synthetase, Indole-3-acetic acid-amido synthetase, acyl acid amido synthetase, and Auxin conjugating enzyme. However, these functions need further experimental verification. The specific situation is in Supplementary Table S10 and Fig. S3. We analyzed expression profiles of large number of *TaGH3* genes, including responses to biotic and abiotic stresses, and 137 different conditions related to growth and developments (Figs. 8 and 9). However, it remained unclear as to how the expression of *TaGH3s* was affected by specific conditions. Gene promoters are essential regulatory factors in modulation of gene expression patterns at both transcriptional and post-transcriptional levels (Fig. 5) [52, 74]. For example, in the salt stress-induced genes *TaGH3-1.8*, *TaGH3-3.13* and *TaGH3-3.15* there were many *cis*-regulatory elements associated with "biological/abiotic stress". Nine to ten G-box motifs ('biotic/abiotic stress' *cis*-regulators) were detected in the promoter regions of *TaGH3-1.4*, *TaGH3-3.3*, and *TaGH3-3.6*, and their expression levels were highly induced by biotic stress, but not by abiotic stress. In summary, *cis*-elements are likely involved in regulation of *TaGH3* expression (Fig. 5). There were many other elements that regulate gene expression, such as DNA methylation and transcription factors in the promoters. More experimental evidence is required to understand the regulatory mechanisms of *cis*-elements affecting TaGH3s.

Tissue-specific expression analysis indicated that the *TaGH3* genes were expressed in leaves and roots, but the expression levels were different. Previous studies have reported that GH3 genes in other species, such as *SlGH3.7*, *MdGH3.4*, *MtGH3.9*, and *OsGH3.2* are highly expressed in roots [13, 36, 90, 91]. Under salt and osmotic stress conditions, most of the present candidate *TaGH3* genes were more highly expressed in roots than in leaves, perhaps indicating that these proteins were more involved in root growth and development. Some TaGH3s were significantly induced in response to abiotic stress treatment, especially in drought-treated conditions with *TaGH3-2.2a*, *TaGH3-3.7*, *TaGH3-3.11*, *TaGH3-3.13* and *TaGH3-3.15* being up-regulated. qRT-PCR analyses showed that *TaGH3* genes were expressed differently in wheat leaves and roots during salt and osmotic stress. Thus, the *TaGH3* family genes appear to play important roles in responding to environmental stress.

Conclusions

We analyzed the common wheat genome to identify and characterize the *GH3* gene family using a range of bioinformatic tools. Thirty six *TaGH3* genes were identified and separated into Groups I–III. Although the proteins varied in length, MW, and Pi, all contained a conserved GH3 structure. Eleven *TuGH3* and 13 *AeGH3* genes were disturbed on seven chromosomes. Twenty-three *TdGH3* genes were disturbed on 10 of 14 chromosomes. Exon/intron and motif analyses showed that the *TaGH3* genes structure and conserved motifs were similar across the group, but the differences between the groups were large. Ka/Ks analysis and interspecies homomorphism analysis indicated that polyploidization was the main reason for the high number of *TaGH3* genes. *Cis*-element analyses indicated that the distribution of *cis*-acting elements of different *TaGH3* genes regulates differences in their functions. Large-scale expression profiling indicated that the *TaGH3* genes were involved in key processes, including plant growth and defense to abiotic and biotic stresses. qRT-PCR analysis revealed that the expression levels of selected *TaGH3* genes were changed by salt and osmotic stress. Our results are the first complete genome analysis to identify the *GH3* genes in wheat and its close relatives. Our findings provide a clearer understanding of the evolutionary relationship of GH3 family genes in wheat, and have potential applications for breeding to enhance stress resistance in wheat.

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Compliance with ethical standards

Conflict of interest All the authors declare no conflict of interest.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

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